IMAGEMAKERS NURSERY AND PRIMARY SCHOOL

Internal Developer Guide

Confidential — for the system developer only

Imparting Wisdom and Morals

Academic session 2026/2027 · LASG Approval No: SLR/14097

Edition: November 2026

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1. Read this first

This internal guide is for the system developer. It assumes you build with AI tools rather than writing code by hand, so each item says what it is, where it lives, and what to ask for next.

> INTERNAL. Do not publish, link or share with the school. It contains no passwords, keys or personal data — keep it that way.

Evidence: source code read on 10 Nov 2026, database functions and policies read live, sign-in pages screenshotted. The test browser could not sign in to this backend (it is a user-managed Supabase project), so no signed-in page was exercised end to end.

2. How the system is built

Layer	What it is	Where
Screens	React + Vite + Tailwind web app	src/pages/{admin,staff,student,parent}
Layouts	Portal shells and navigation	src/components/layout/*Layout.tsx
Sign-in & roles	Supabase Auth + user_roles table	src/hooks/useAuth.tsx, ProtectedRoute
Database	Supabase Postgres with row-level security	supabase/migrations
Server code	Edge functions	supabase/functions
School rules	Shared constants (clock-in, weights, specialist subjects)	src/lib/schoolConfig.ts

3. Routes by portal

Portal	Routes
Public	/login, /parent-access, /apply, /s/:token (QR), /p/:token, /verify, /reset-password
Admin	/admin, students, staff, users, bulk-upload, register-import, id-cards, approvals, reports, progression, cbt, oversight, finance, fees, financial, announcements, communication, policies, documents, analytics, predictive, wellbeing, behavioral, library, inventory, visitors, substitutions, complaints, settings, activity, transport, certificates
Staff	/staff, classes, attendance, gradebook, report-card, reports, cbt, assignments, lesson-plans, students, admissions, messages, forum, leave, wall, profile
Student	/student, grades, reports, report-card, transcript, schedule, attendance, fees, announcements, exams, exams/:id, homework, learning, library, calendar, complaints, resources, wall, profile, settings
Parent	/parent, grades, attendance, fees, messages, forum, wall, profile, transport

4. Sign-in and roles

- Roles live in **user_roles** (student, teacher, admin, parent, driver). Users cannot change their own role.
- Pupils: QR scan calls the **qr-student-login** function, which creates the pupil's login on first scan and links students.user_id.
- Parents: **parent-login** function checks Parent ID + access code.

- Admin preview: useAuth gives an admin without own records a real class teacher (largest class) and a pupil from that class; parent pages fall back to that pupil (src/lib/adminPreview.ts). Choice is stored in the browser as im_preview_teacher_id / im_preview_student_id. Read-only display; writes still run as the admin.
- Key database helpers: has_role, is_staff, is_super_admin, has_admin_permission, can_manage_student, can_view_student, teaches_class.

5. Data and security rules

Area	Tables	Who can read / write (live policy)
Pupils & results	students, grades, term_results	Read: can_view_student (staff for the pupil, the pupil, linked parent). Write: can_manage_student
Finance	invoices, invoice_lines, receipts, receipt_allocations	Read: can_view_student. Write: has_admin_permission(can_manage_fees). Allocations written only by trigger
Audit	finance_audit	Insert/read only; update and delete denied
Progression	student_progression_history	Admin manage; pupil reads own. Parents: no policy
Timetable	schedules	All signed-in read; staff write
Lesson notes	lesson_plans	Staff read/write
Staff clock-in	staff_attendance	Staff read; own insert/update or admin
CBT	exams, exam_questions, exam_submissions	Questions staff-only; pupils use student_exam_questions (no answer key) and submit_exam (server scoring)
Admin permissions	admin_permissions, admin_role_templates	Super admin manages

6. Key flows

Grades

Trigger **compute_grade_fields** sets total = CA + exam, then remark (90+ Outstanding, 70+ Excellent, 60+ Very Good, 50+ Good, 40+ Fair, else Poor) — matches school bands. It ALSO sets letter_grade A–F (90/80/70/60/50) which the school has not approved.

Invoices

generate_term_invoices(term, class, due): requires can_manage_fees; one invoice per active pupil per term (no duplicates); deletes and re-inserts only lines of kind 'fee' from active catalogue items. Optional items are expected to come from student_fee_selections.
record_manual_payment creates a receipt; trigger allocate_receipt_waterfall spreads it over lines by priority; recalc_invoice updates totals.

Progression

record_student_progression: admin only, locks the pupil row, validates decision/class rules, updates class and inserts history in one transaction (atomic).

7. Implementation status

Feature	Status	Frontend	Backend	Tests	Confidence	Remaining work
Role sign-in & portals	Implemented, unverified	useAuth, layouts	user_roles, has_role	None	Medium	Signed-in click-through per role
Admin preview identity	Implemented, unverified	useAuth, adminPreview.ts	Uses existing staff read access	None	Medium	Add a picker to choose teacher/pupil
QR pupil login	Implemented, verified	StudentPortalEntry	qr-student-login	Manual call OK	High	Reprint old ID cards
Class-teacher scoping	Implemented, unverified	StaffClasses, useAssignedClass	classes.class_teacher_id	None	Medium	Test with a teacher login
40/60 marks + remarks	Implemented, unverified	StaffGradebook	compute_grade_fields	None	Medium	Remove A–F letter; confirm entry maxima
Specialist A–E grades	Not implemented	Config only	None	None	High	Entry screen, report-card row, boundaries from school
Result publishing	Partially implemented	reportCards.ts	term_results.is_published	None	Low	Confirm pupil/parent views filter unpublished
Historical reports	Partially implemented	StudentReportCard	Data kept per term	None	Low	Session/term selector for pupil and parent
Progression	Implemented, unverified	AdminProgressionPage	record_student_progression	None	Medium	Parent read policy; notifications
Fee catalogue mandatory/optional	Implemented, unverified	AdminFeesPage	fee_items, student_fee_selections	None	Medium	Verify optional-only billing
Textbooks / notebooks	Not implemented	—	—	None	High	Needs school price list
Invoices & re-generation	Implemented, unverified	AdminFinancePage	generate_term_invoices	None	Medium	Test re-run keeps discounts
Manual payments & receipts	Implemented, unverified	Finance, ReceiptDocument	record_manual_payment	None	Medium	Duplicate-reference guard
Paystack	Not implemented	—	—	None	High	Merchant account, init + webhook functions
Principal/Bursar separation	Partially implemented	Menu hiding	Only fees gated	None	High	Policy changes per permission
Staff clock-in	Implemented, unverified	Staff dashboard, Oversight	staff_attendance	None	Medium	Radius approval
Teacher timetable editing	Not implemented	View only	Staff may write	None	High	Editor screen
Lesson-start signal	Not implemented	—	—	None	High	Build
Lesson-note approval	Not implemented	View only	status column	None	Medium	Approve/return actions

Feature	Status	Frontend	Backend	Tests	Confidence	Remaining work
Online CBT	Implemented, unverified	StaffCBT, TakeExam	student_exam_questions, submit_exam	None	Medium	Timed end-to-end test
Offline Windows CBT	Not implemented	Spec docs only	—	None	High	Build per docs/CBT_DESKTOP_*
AI helper	Implemented, unverified	AIChatWidget	ai-assistant (JWT required)	401 check OK	Medium	Image input; moderation tests
Notifications permission	Not implemented	No permission request	notifications table	None	High	Request + push
Wall & announcements	Implemented, unverified	Wall pages	wall_*, announcements	None	Medium	Post/read test per role
Analytics / predictive	Implemented, unverified	Admin analytics pages	Live tables	None	Low	Recheck once real data exists

8. Gaps and acceptance tests

Gap	Confirmed	Work required	Acceptance test
Admin permission flags	Only fees enforced in database	Add has_admin_permission checks to grades, students, staff policies	Bursar account cannot read grades via the app
Parent progression read	No parent policy	Add SELECT policy using parent_student_links	Linked parent sees history; unlinked does not
Class-change notification	None	Insert notifications in record_student_progression	Pupil and parent receive notice
A–F letter grade	Trigger still sets it	Stop writing/displaying letter_grade for core subjects	Report card shows remark only
Optional fees	Depends on selections join	Verify the function filters non-mandatory by selection	Unselected uniform absent from invoice
Invoice re-run	Only kind='fee' lines replaced	Confirm discount lines survive	Re-run keeps discount and payments
Payment idempotency	No reference uniqueness seen	Unique (method, reference) guard	Same reference twice is refused
Paystack	No code	Init + verify + webhook functions	Test payment produces one receipt
Unpublished results	Not confirmed in views	Filter is_published in pupil/parent queries	Draft result invisible to parent
Notifications	No permission prompt	Request on dashboard, add push	Prompt appears once on phone

9. Contradictions with older documents

- FUNCTIONAL_SPEC.md describes A–F as official; the school approved descriptive remarks.
- Older notes mention a 7:45 a.m. cutoff; config and school decision now say 7:30.
- Older notes treat 400 m as policy; it is unconfirmed.

- Transport and driver features exist but school transport is outsourced.

10. Information still needed from the school

- Textbook titles, classes, quantities and prices; 40/60/80-leaf notebook prices.
- Whether clubs are compulsory and whether one club per pupil.
- Specialist A–E boundaries, entry method and whether they count to the 50% average.
- Whether 50% is per term or per session.
- Approval of the 400 m radius and any grace period.
- Official admission numbers for all pupils.
- Paystack registration under the school's CAC with the bursar as contact.

11. Do not yet describe as finished

- Online payments
- Principal/Bursar separation
- Specialist A–E grading
- Teacher timetable builder and lesson-start signal
- Lesson-note approval
- Phone notifications
- Offline Windows CBT
- AI image reading
- Parent view of promotion history and older report cards

12. Rebuilding these guides

Edit docs/guides-source/build_guides.py and run: python3 docs/guides-source/build_guides.py. Screenshots live in docs/guides-source/screenshots.